Class 03 Handout – LP Sensitivity Analysis

A company is considering creating a new diet drink using 4 different ingredients. Three chemicals X, Y, and Z will be used in the mix and the amount of each ingredient in each chemical is given. With the objective of minimizing the cost, how many ounces of each ingredient should be used? There is a minimum daily dosage of 36 and maximum of 40 for the drink containing all four ingredients.

	Ingredient				
	A	B	C	D	Requirement
Chemical X	3	4	8	10	At least 280 units
Chemical Y	5	3	6	6	At least 200 units
Chemical Z	10	25	20	40	At most 1,050 units
Cost/ounce	$40	$20	$60	$30	

The LP model and the Solver sensitivity report are provided:

$$Min\ 40A + 20B + 60C + 30D$$

$$A + B + C + D \geq 36 \quad \text{(Daily Dose Min)}$$

$$3A + 4B + 8C + 10D \geq 280 \quad \text{(X Requirement)}$$

$$5A + 3B + 6C + 6D \geq 200 \quad \text{(Y Requirement)}$$

$$10A + 25B + 20C + 40D \leq 1{,}050 \quad \text{(Z Requirement)}$$

$$A + B + C + D \leq 40 \quad \text{(Daily Dose Max)}$$

$$A,\ B,\ C,\ D \geq 0$$

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C3	Ounces A	10.25	0	40	6.11111111	25
D3	Ounces B	0	6.875	20	1E+30	6.875
E3	Ounces C	4.125	0	60	150	7.33333333
F3	Ounces D	21.63	0	30	8.46153846	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
G6	Daily Dose Min	36	37.5	36	4	1.27777778
G7	X Req	280	8.75	280	41	11
G8	Y Req	205.8	0	200	5.75	1E+30
G9	Z Req	1050	-2.375	1050	47.1428571	346
G10	Daily Dose Max	36	0	40	1E+30	4

1- Using the information in the report:
   a. What is the optimal solution? What is the optimal value for the objective function?

   b. Which constraints are binding? Which ones are not binding? Identify the amount of slack/surplus.

Variable Cells

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D3	Ounces B	0	6.875	20	1E+30	6.875
E3	Ounces C	4.125	0	60	150	7.33333333
F3	Ounces D	21.63	0	30	8.46153846	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
G6	Daily Dose	36	37.5	36	4	1.27778
G7	X Req	280	8.75	280	41	11
G8	Y Req	205.75	0	200	5.75	1E+30
G9	Z Req	1050	-2.375	1050	47.1429	346
G10	Daily Dose	36	0	40	1E+30	4

2- What is the impact on cost if we increase X requirement to 290? What if we decrease to 275? What if we decrease to 250?

3- What is the impact on cost if we increase Z requirement to 1070? What about decreasing to 1020?

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D3	Ounces B	0	6.875	20	1E+30	6.875
E3	Ounces C	4.125	0	60	150	7.33333333
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4- What is the impact on cost if we increase the daily dosage maximum to 100?

5- Suppose we have a new ingredient "E" which costs $30, and contains 5, 7, and 10 unit of chemicals X, Y, and Z, respectively. Is it optimal to use it in the drink? What if its cost is $70?

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D3	Ounces B	0	6.875	20	1E+30	6.875
E3	Ounces C	4.125	0	60	150	7.33333333
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Constraints

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6- It is not profitable to use ingredient B in the production.
   a. What is the impact on cost if we insist on using 1 ounce of ingredient B to make the drink?

   b. How much cheaper the price of ingredient B must be to make it profitable to use?

7- There is uncertainty about the cost of ingredient C. How much it can change without changing the solution?